

Step 0: Project folder structure

powershell

mkdir backend, frontend

Step 1: Install Ollama (LLM runtime) — do this FIRST, it takes longest (downloads)

Download and install from <https://ollama.com/download/windows> (GUI installer, not a terminal command on Windows).

Once installed, open a **new terminal** and pull your models:

powershell

ollama pull llama3.3

ollama pull phi4-mini

ollama pull nomic-embed-text

Verify it's running:

powershell

ollama list

Step 2: Backend setup (Python)

powershell

cd backend

python -m venv venv

.\venv\Scripts\Activate.ps1

If activation is blocked by PowerShell execution policy, run this once first:

powershell

Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser

Now install all backend libraries in one go:

powershell

pip install fastapi uvicorn pydantic-settings

pip install drain3

pip install scikit-learn pandas

pip install prophet

pip install langchain langchain-community

```
pip install ollama
```

```
pip install sqlalchemy psycopg2-binary
```

```
pip install chromadb
```

```
pip install sentence-transformers
```

```
pip install faker
```

Save your dependencies for teammates:

```
powershell
```

```
pip freeze > requirements.txt
```

Step 3: PostgreSQL setup

Download installer from <https://www.postgresql.org/download/windows/> (GUI installer — set a password for the postgres user during setup, remember it).

After install, create your project database (open **SQL Shell (psql)** from Start Menu):

```
sql
```

```
CREATE DATABASE sih2026;
```

Step 4: Frontend setup (React)

```
powershell
```

```
cd ..
```

```
npm create vite@latest frontend -- --template react
```

```
cd frontend
```

```
npm install
```

Now add all frontend libraries:

```
powershell
```

```
npm install -D tailwindcss postcss autoprefixer
```

```
npm install recharts
```

```
npm install react-router-dom
```

```
npm install lucide-react
```

Initialize Tailwind:

```
powershell
```

```
npx tailwindcss init -p
```

Step 5: Verify everything works

Test backend:

```
powershell
```

```
cd ../backend
```

```
uvicorn main:app --reload
```

(This will fail right now since main.py doesn't exist yet — that's expected, just confirming FastAPI/uvicorn installed correctly. You'll see a "module not found" error, not an installation error.)

Test frontend:

```
powershell
```

```
cd ../frontend
```

```
npm run dev
```

This should open a working blank Vite+React app in your browser — if you see that, frontend setup is confirmed working.

Test Ollama:

```
powershell
```

```
ollama run llama3.3 "Say hello in one sentence"
```

Should return a response with no internet needed.

Order of operations for today

1. Install Ollama first (biggest download, run in background while you do other steps)
2. Install PostgreSQL (second biggest, GUI installer)
3. Backend Python packages while those download
4. Frontend npm packages
5. Run the 3 verification tests above